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# RNA-SEQ ANALYSIS MADE ~~EASY~~ Even Easier!


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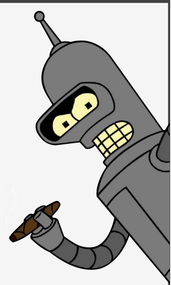
Try me!



GitHub
https://github.com/zavolanlab/zarp

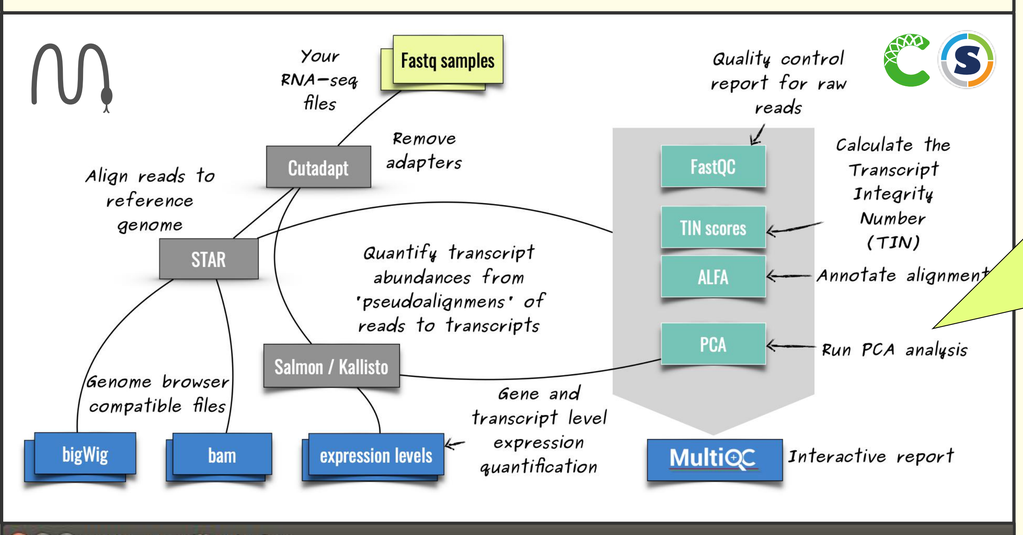
ZAVOLAB · BIOZENTRUM The Center for Molecular Life Sciences · University of Basel · SIB Swiss Institute of Bioinformatics

ZARP

NOW WITH CLI!

## ...previously in ZARP Adventures™

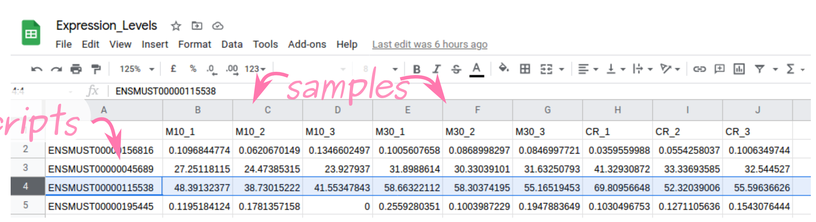
### What does ZARP do?




IGV — Mozilla Firefox | IGV | ZARP: MultiQC Report | Snakemake Report

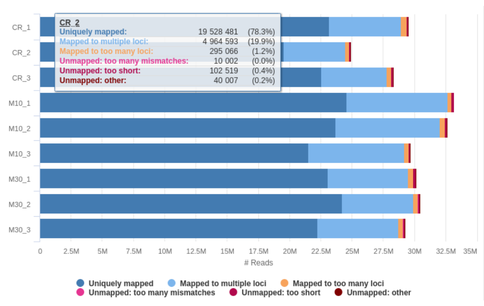
### What does ZARP give?

explore the results in your browser!

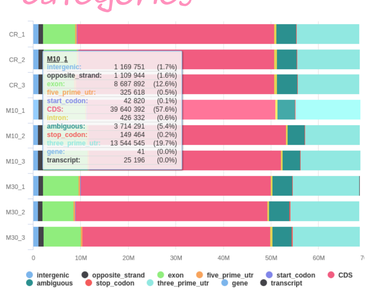



Showing 4 of MANY plots...

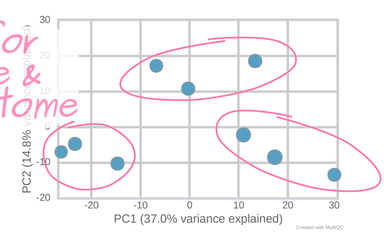
mapping statistics

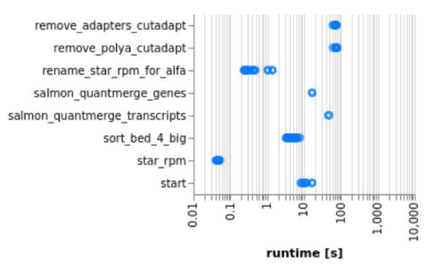


genomic categories



PCAs for genome & transcriptome



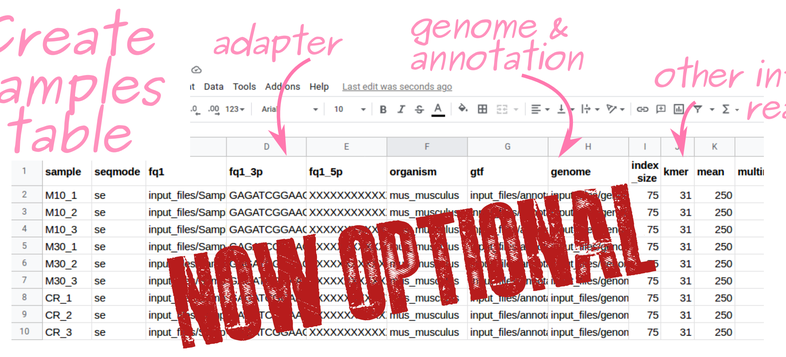




runtime statistics

### What does ZARP need?

Create samples table

adapter | genome & annotation | other info about reads



NOW OPTIONAL

## ...last week in The Lab

Whoa, I analyzed 358 RNA-seq samples all by myself. Just ZARP'd them. Lit af!

Meeeh, ZARP! Such a pain to fill out those stupid sample tables... Why do we have bioinformaticians? I'll just keep sending it their way...

And wait months until they'll get to it? Just use ZARP-cli! Thank me later...

Launches ZARP!

ZARP-cli

Fetches and manages **genome resources**!

powered by: https://github.com/vanheeringen-lab/genomepy

support for hundreds of organisms!

great for batch analyses!

Infers sample **metadata**:
- Organism
- 3' adapters
- Read orientation

EXPERIMENTAL

powered by: https://github.com/zavolanlab/htsinfer

Manages your ZARP default **configuration**. Set up once and profit!

set it up for yourself, or for your lab

It's THAT simple!

```
> zarp sample_1.fq.gz
```

OR

```
> zarp SRR1234 SRR1235
```

OR

```
> zarp table:samples.tsv
```

Runs ZARP on
- local files
- SRA accesions
- sample tables

even incomplete ones!

## Why ZARP?

- Easy to set up
- Easy to run
- Explore results in browser
- Fast, free, fun§

- Configurable
- Extendable
- Reproducible
- Fast, free, fun§

§ depending on personal disposition

To be continued...